

Application of pumping lemma for regular sets

Note: Pumping lemma is a technique to identify the class of language for finite automata (FA).

Example-1: Show that a language $L = \{a^n b^n : n > 0\}$ is not regular.

Solution:

Let L be a regular language and any string $w = a^p b^p$, $w \in L$.

Now, according to the pumping lemma of regular set, 'w' can be decomposed into x , y , and z as:

$$\begin{aligned}x &= a^q \\y &= a^r \quad (r > 0) \\z &= a^{p-(q+r)} b^p, \text{ where } w = xyz \in L\end{aligned}$$

Now, using the property of pumping lemma, we have

$$xy^2z = a^q a^{(r)*2} a^{p-(q+r)} b^p = a^q a^{2r} a^{p-(q+r)} b^p = a^{p+r} b^p \neq a^p b^p \text{ or } xy^2z \notin L$$

Here, our assumption $w \in L$ contradict with our result obtained. Hence, language $L = \{a^n b^n : n > 0\}$ is not regular.

Note: In the decomposition of the string, we must decompose the first symbol for the first and second string and then put all remaining parts on the final string.

Example-2: Show that a language $L = \{xx : x \in (0, 1)^*\}$ is not regular.

Solution:

Let L be a regular language and $x = a^n b^n$ then any string $w = a^n b^n a^n b^n$, $w \in L$.

Now, according to the pumping lemma of regular set, 'w' can be decomposed into the strings r , s , and t as:

$$\begin{aligned}r &= a^l \\s &= a^m \quad (r > 0) \\t &= a^{n-(l+m)} b^n a^n b^n, \text{ where } w = rst \in L\end{aligned}$$

Now, using the property of pumping lemma, we have

$$rs^2t = a^l a^{(m)*2} a^{n-(l+m)} b^n a^n b^n = a^l a^{2m} a^{n-(l+m)} b^n a^n b^n = a^{n+m} b^n a^n b^n \neq a^n b^n a^n b^n \text{ or } rs^2t \notin L$$

Here, our assumption $w \in L$ contradict with our result obtained. Hence, language $L = \{xx : x \in (0, 1)^*\}$ is not regular.

Example-3: Show that a language $L = \{a^{i^2} : i \geq 1\}$ is not regular.

Solution:

Here, we have to show that the pumping lemma does not hold for L . Thus, consider any pumping number $p \geq 1$.

Now, choose $w = a^{p^2}$

Now, according to the pumping lemma of regular set, $w = xyz$ can be decompose such that $|y| > 0$ and $|xy| \leq p$. That is:

$$\begin{aligned}x &= a^m \\y &= a^n \quad (n > 0) \\z &= a^{p^2-(m+n)}\end{aligned}$$

Now, using the property of pumping lemma, we have

$$xy^2z = a^m a^{(n)*2} a^{p^2-(m+n)} = a^m a^{2n} a^{p^2-(m+n)} = a^{p^2+n} \neq a^{p^2} \text{ or } xy^2z \notin L$$

Here, our assumption $w \in L$ contradict with our result obtained. Hence, language $L = \{a^{i^2} : i \geq 1\}$ is not regular.

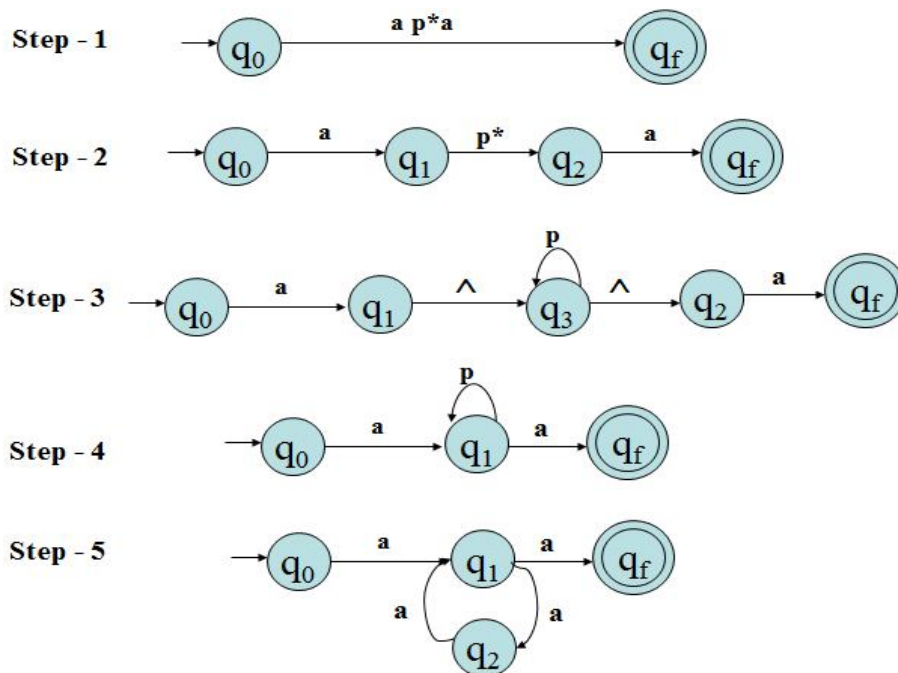
Example-4: Show that a language $L = \{a^{2n} : n \geq 1\}$ is regular.

Solution:

Here, we can write, $L = a^{2n} = a(a^2)^i a$, $i \geq 0$ [As $i = 0$, $L = aa$ and $i = 1$, $L = a(a^2)^1 a$]

Now, let $a^2 = p$

Hence, $L = ap^*a$, which can be represented by finite automata as given below:



Hence, L is a regular language as it is accepted by finite automata.